

Intestinal Permeability:

Manifestations – Mediators – Mitigators



SWATHI RAO, PA-C

APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE

Disclosures

Swathi Rao, PA-C, disclosed no relevant financial relationships with any ineligible companies.

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Learning Objectives

At the conclusion of this segment, successful participants will be able to:

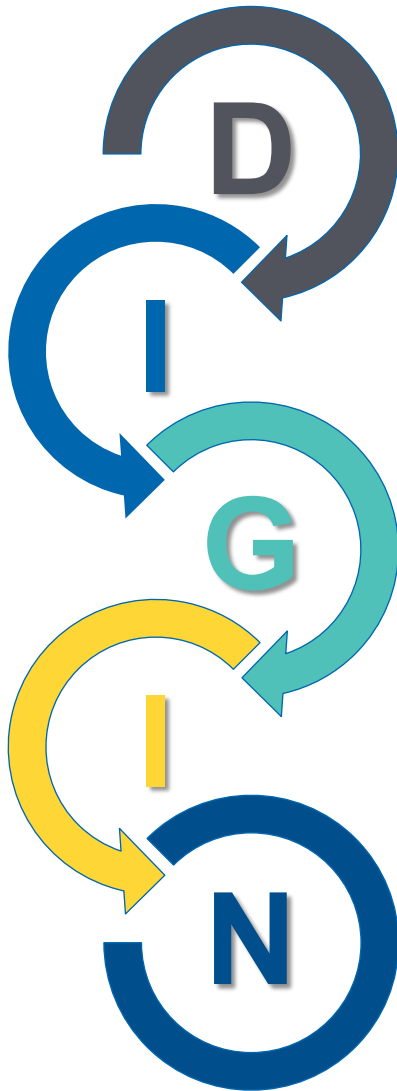
1. Define key terms related to intestinal permeability.
2. Recall the most common clinical findings, antecedents, triggers, and mediators of intestinal permeability.

IFM Clinical Skills and Performance Objectives

THIS SEGMENT WILL ENABLE PARTICIPANTS TO DEVELOP & APPLY THESE SKILLS IN THEIR PRACTICE:

1. Identify the antecedents, triggers, mediators, and clinical findings of increased intestinal permeability and document them on the patient's Timeline and Matrix.
2. Identify extraintestinal diseases and dysfunctions that may be mediated by increased intestinal permeability.

The DIGIN Mnemonic of Gut Dysfunctions



Digestion/Absorption

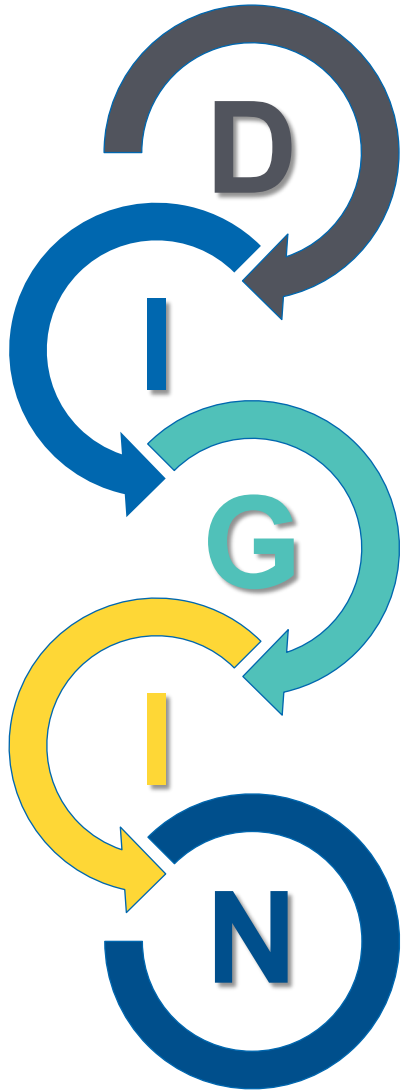
Intestinal Permeability

Gut Microbiota/Dysbiosis

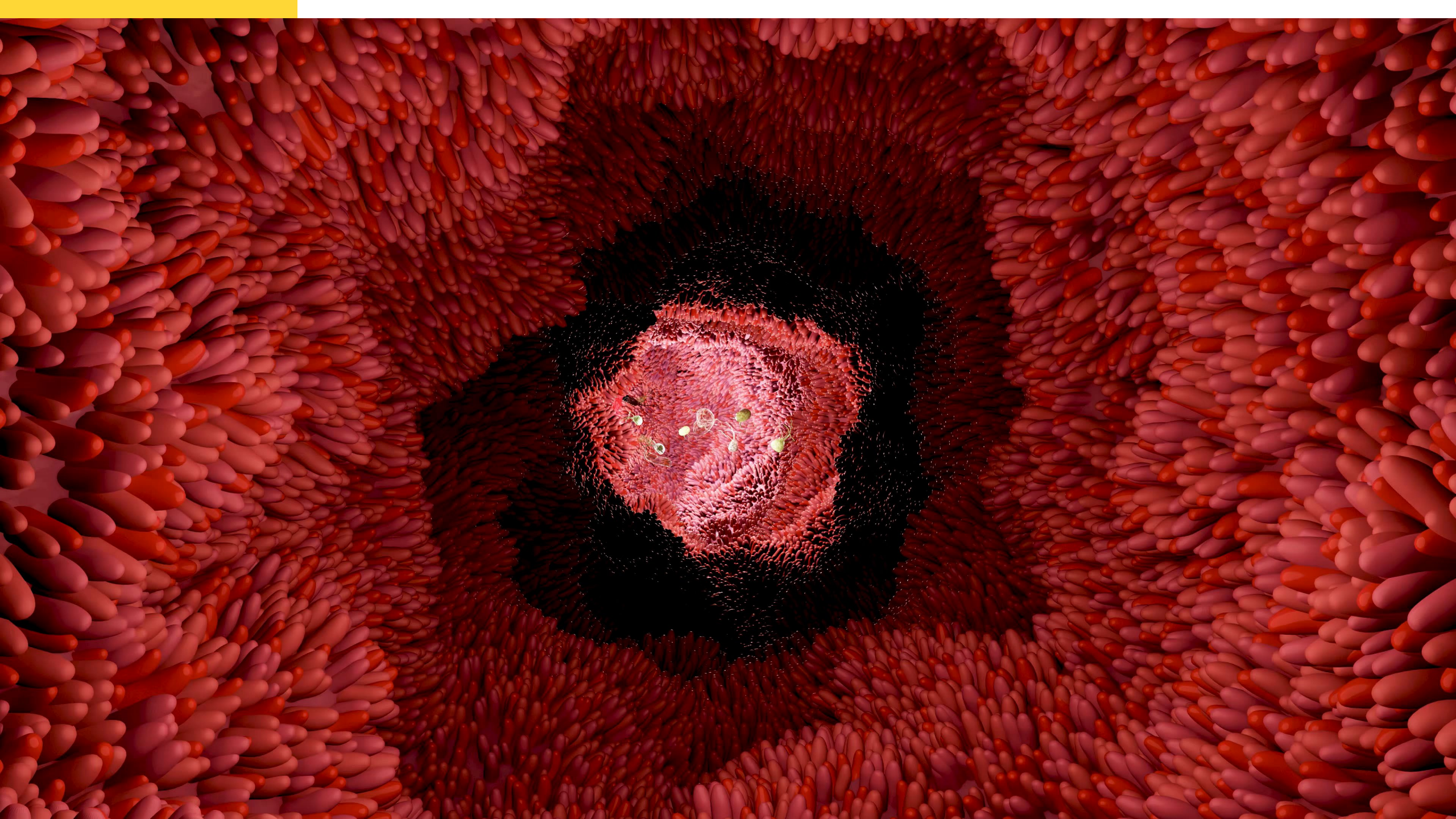
Inflammation/Immune

Nervous System

The DIGIN Mnemonic of Gut Dysfunctions



Intestinal Permeability



Long Description: Intestinal Barrier

- The structure and function of the intestinal barrier are essential to absorption and immune function

Clinical Focus

- Definitions of key terms related to increased intestinal permeability.



Terminology

Intestinal permeability is always present to some extent and is necessary for normal physiological function.

While some intestinal permeability is normal, dysfunction occurs when the protective barrier of cells are damaged and no longer tightly connected. This is referred to as “increased intestinal permeability.”

Increased intestinal permeability (also referred to as a “leaky gut”) allows harmful substances and partially digested food to enter the submucosa and trigger a dysregulated immune response that may lead to systemic and chronic inflammation and disease.

Terminology

Tight Junctions: Structures made of proteins that seal the epithelial cells in the intestinal lining to maintain the integrity of the barrier and regulate the movement of molecules across it.

Zonulin: A protein that modulates tight junctions, thereby regulating the movement of substances across the intestinal barrier.

Terminology

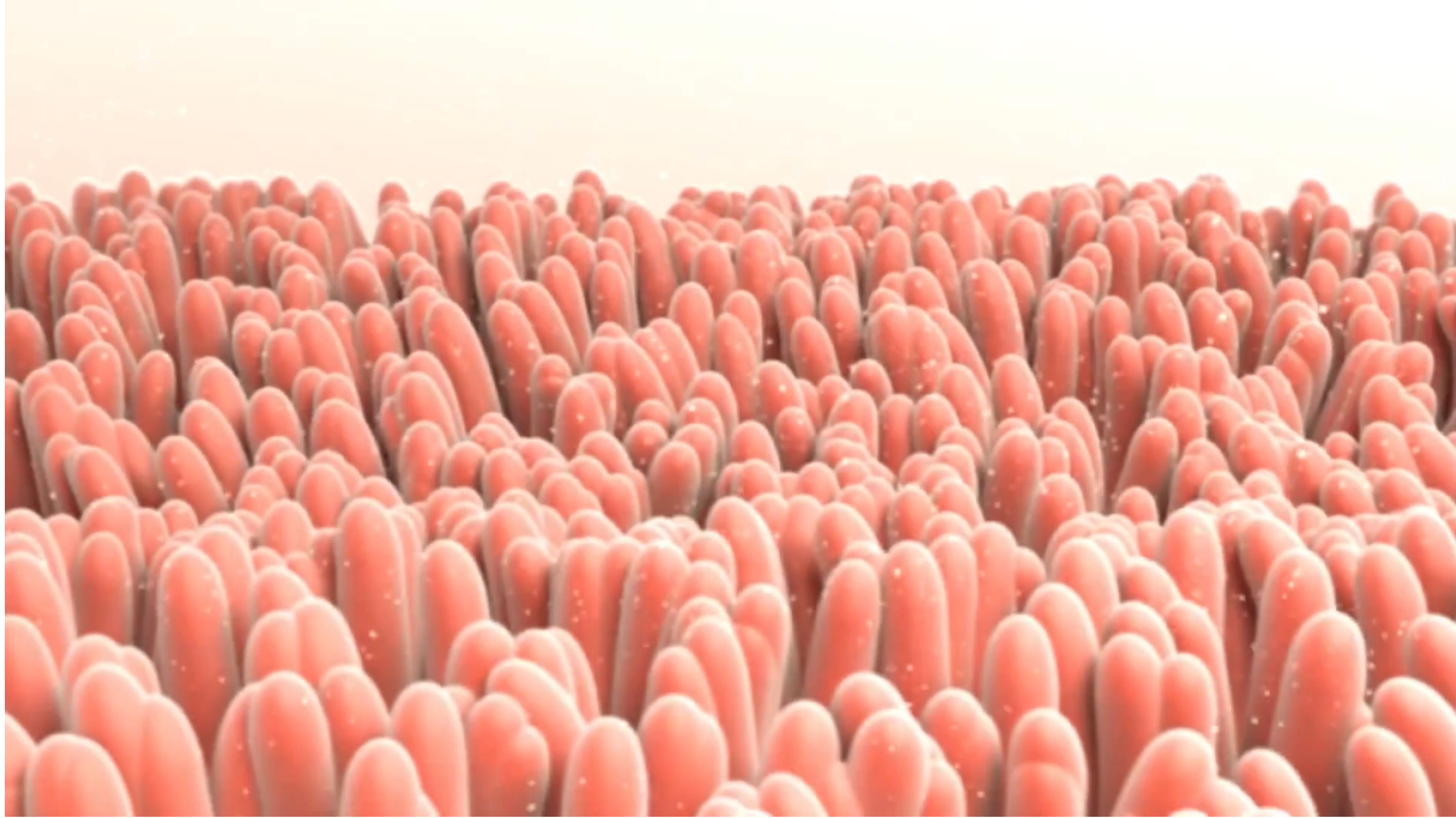
Translocation: Movement of substances from the gut lumen across the intestinal lining into the bloodstream.

Lipopolysaccharides (LPS)/Endotoxins: Components of the outer membrane of gram-negative bacteria, most often produced by dysbiotic strains. When increased intestinal permeability is present, endotoxins can enter the bloodstream and mediate an inflammatory response.

Clinical Focus

- Normal physiology of the gut lining.
- Pathological mechanisms of increased intestinal permeability.

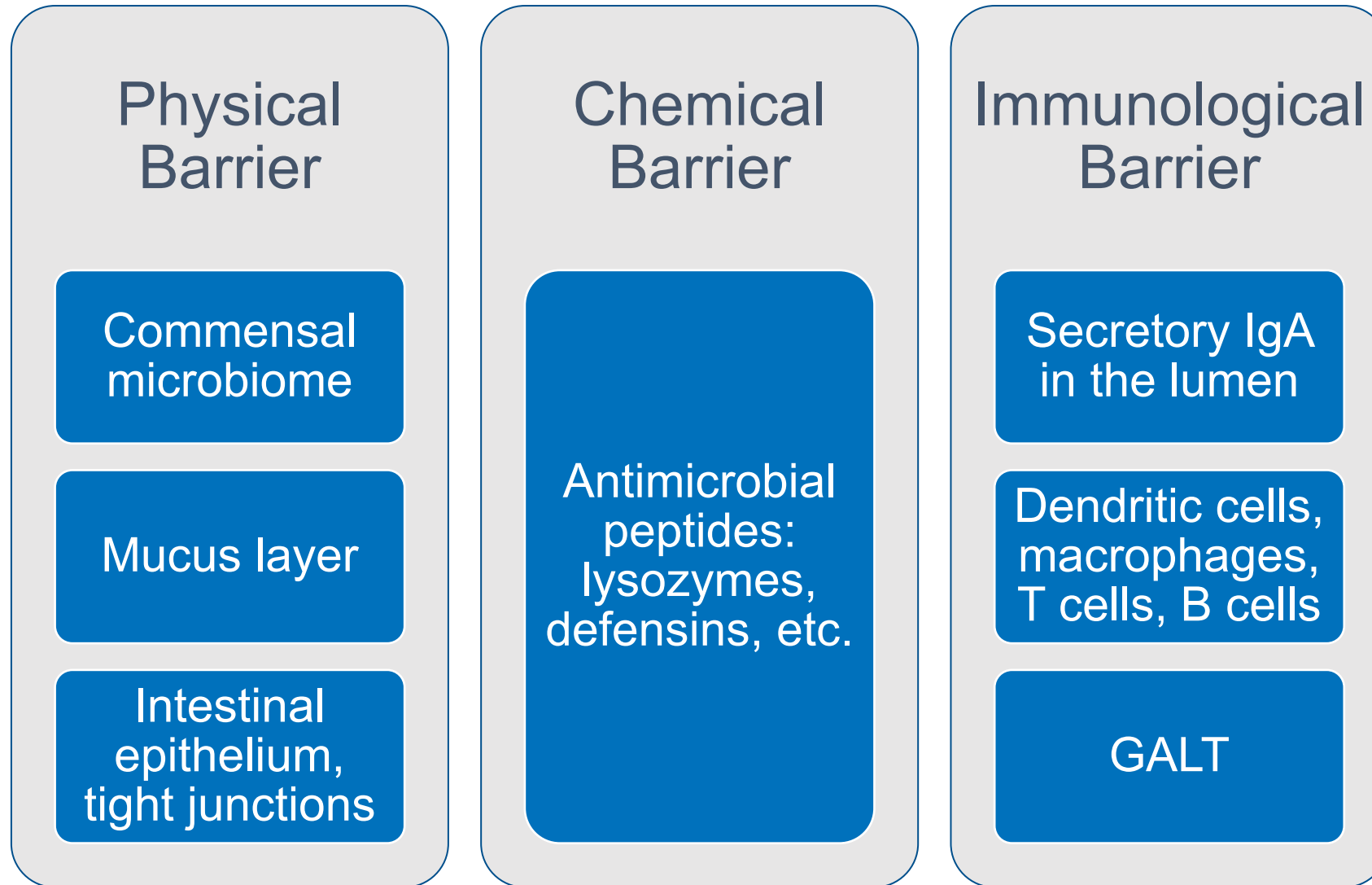




Long Description: Intestinal Villi Video

- Intestinal villi are highly vascularized and contain lymphatics to carry nutrients into systemic circulation.

Components of the Intestinal Barrier



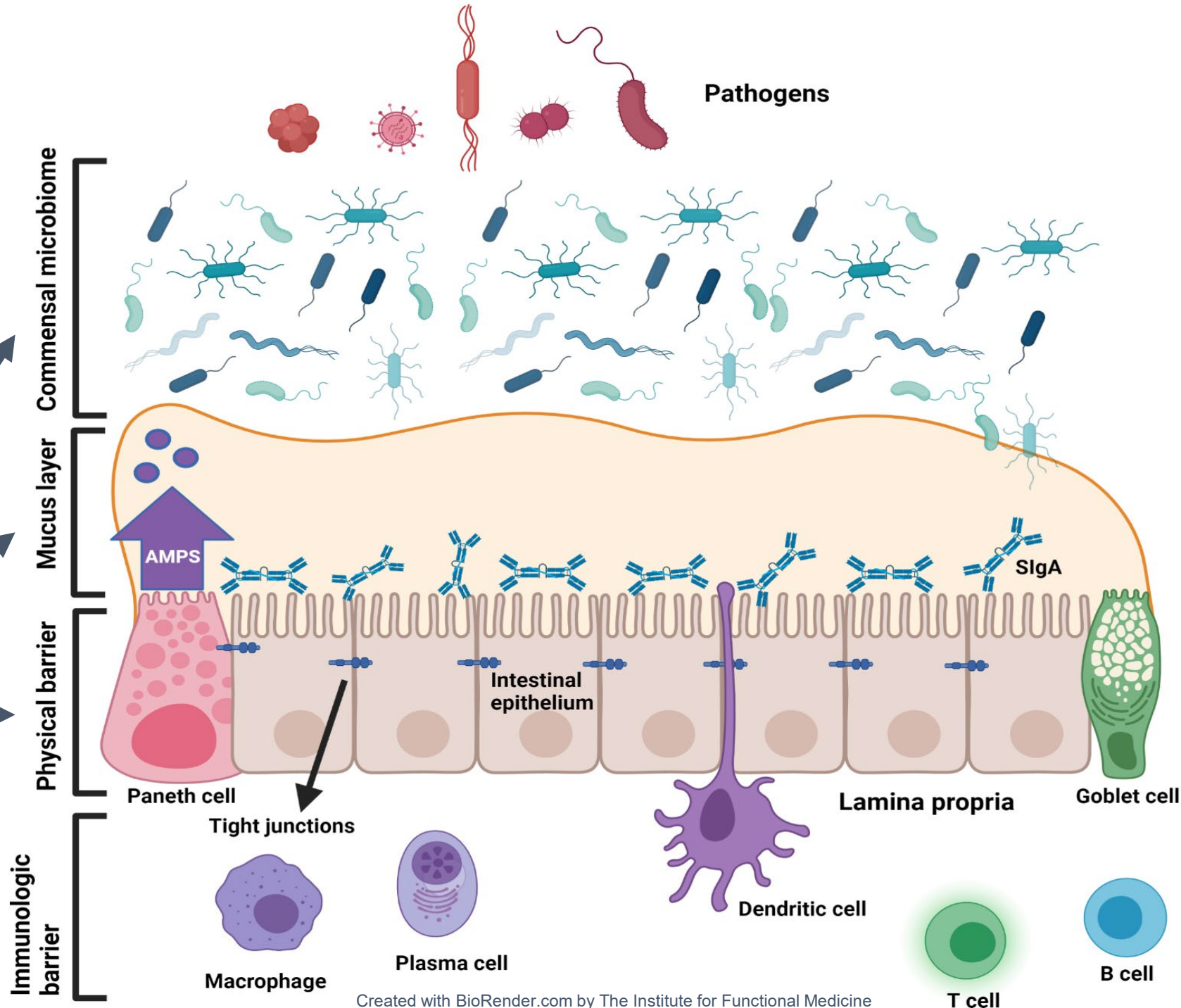
Chemical Barriers:

Antimicrobial peptides (AMPs): lysozymes, defensins, etc.

Physical Barriers

Immunological Barriers

See: "References: Components of the Intestinal Barrier"

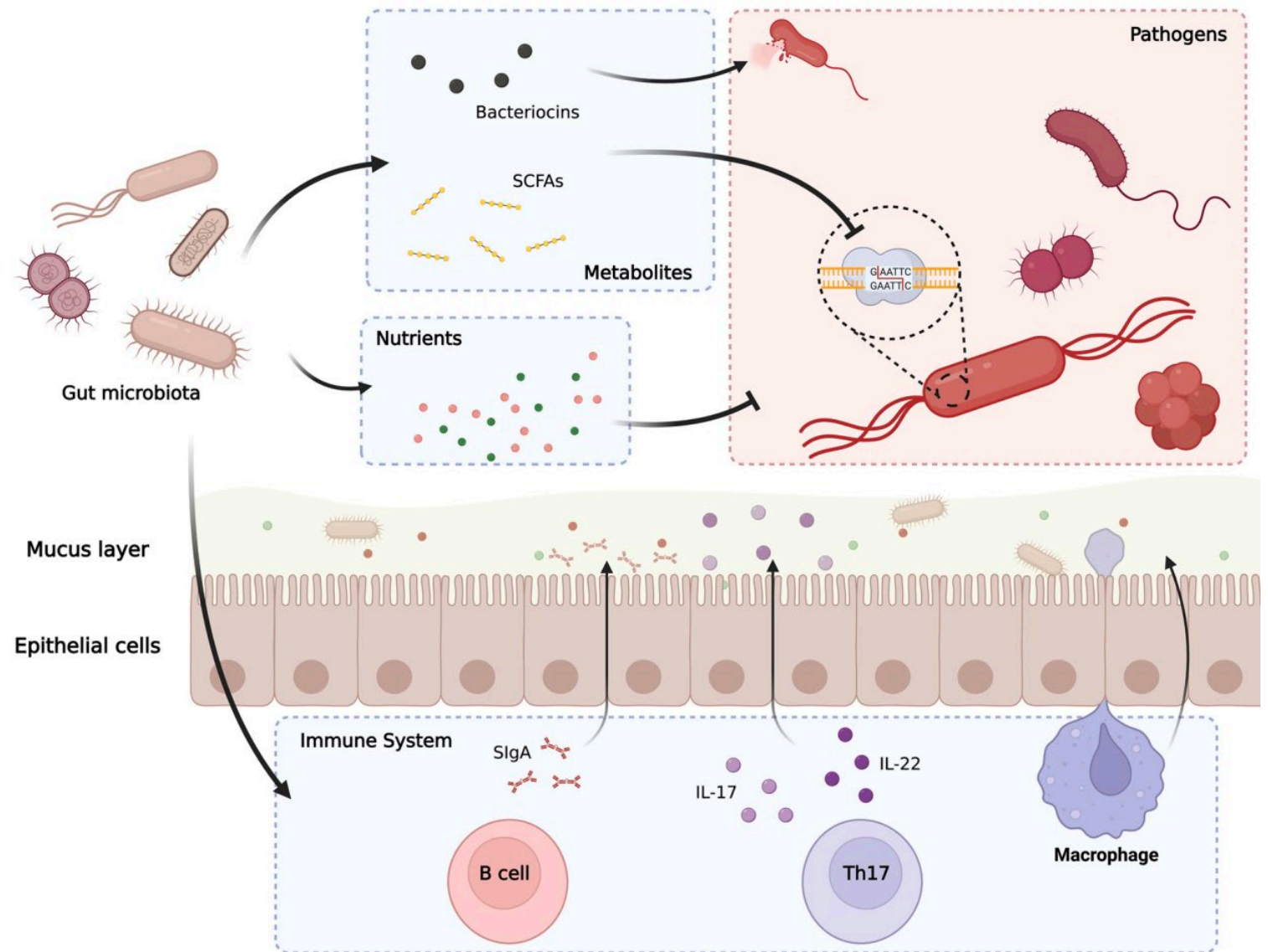


Long Description: Intestinal Barriers

- The intestinal barrier is comprised of physical, chemical, and immunological barriers. The physical barrier includes the commensal microbiome in the intestinal lumen, a mucus layer, and an underlying monolayer of intestinal epithelium joined by tight junctions interspaced with Paneth and Goblet cells. Paneth cells secrete antimicrobial peptides such as lysozymes and defensins, which are part of the chemical barrier. Beneath the epithelium, the lamina propria contains the immunological barrier composed of macrophages, plasma cells, dendritic cells, T cells, and B cells. Secretory IgA is a component of the immunological barrier secreted into the intestinal lumen.

MICROBIOME FUNCTIONS

- Metabolic benefit
- Release bacteriocins, which are antimicrobial
- Inhibit colonization of pathogenic microbes
- Produces SCFAs to support the mucus layer
- Intestinal barrier protection
- Immune function regulation

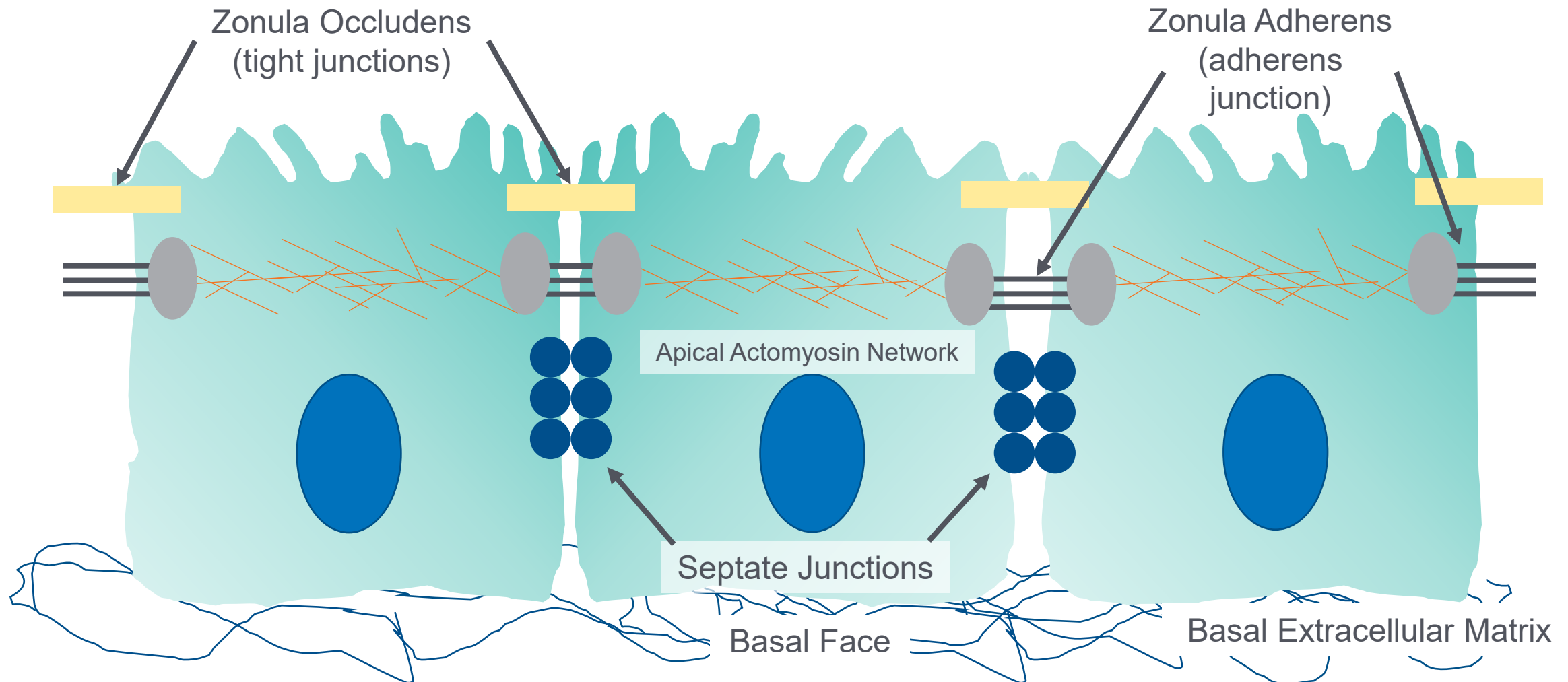


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COMPONENTS OF THE INTESTINAL BARRIER

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Components of Normal Intestinal Permeability



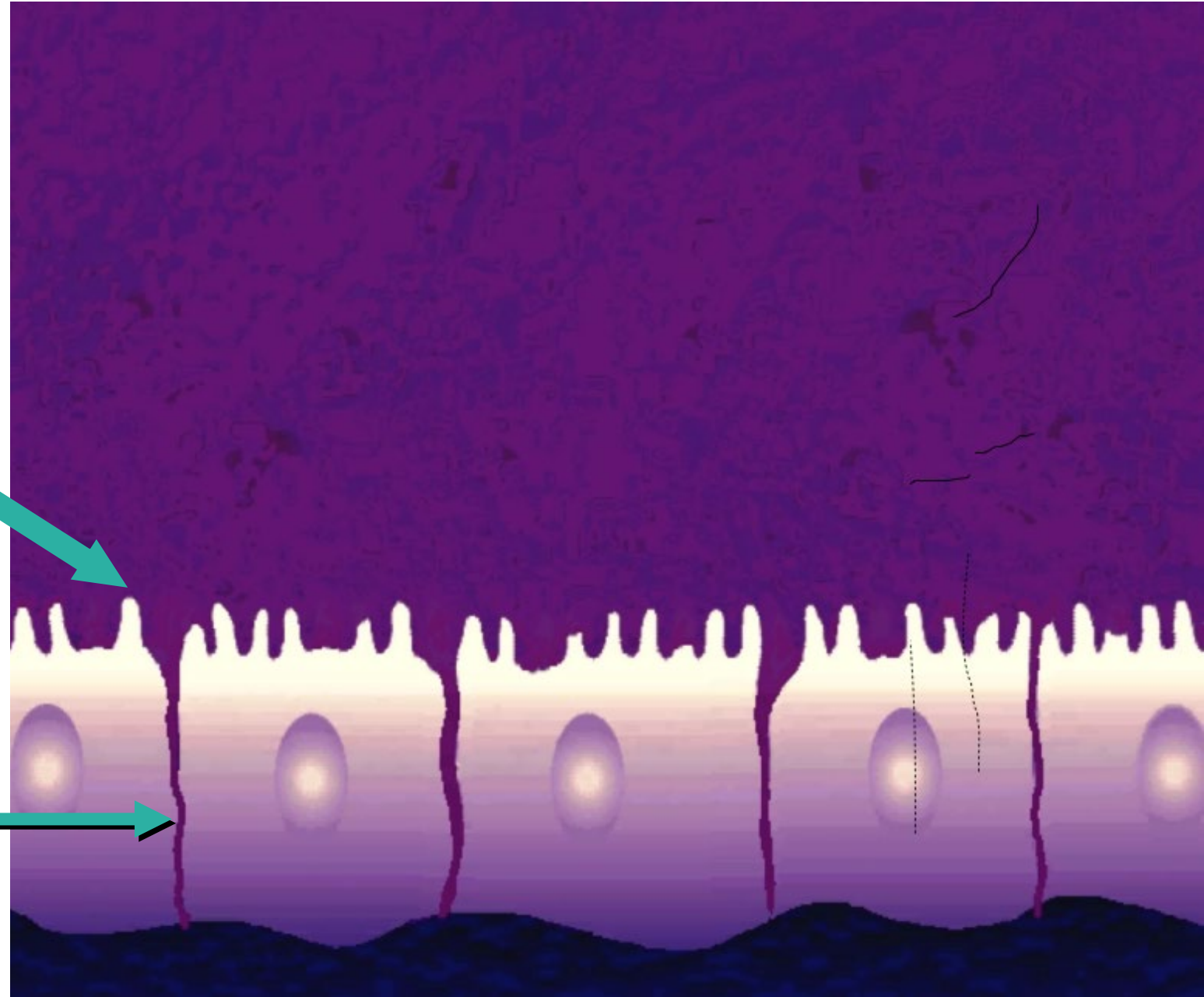
Long Description: Components of Normal Intestinal Permeability

- Epithelial cells are connected by zonula occludens, or tight junctions, at the apical surface. Beneath them are the zonula adherens, or the adherens junctions, connecting to the apical actomyosin network.

Leaky Gut & Malabsorption

**Damaged
Villi/Poor
Absorption**

**Damaged
Cell
Junctions**



Long Description: Leaky Gut and Malabsorption

- Damaged villi and damaged cell junctions permit intestinal permeability and malabsorption.

NSAIDs, dysbiosis, enterotoxins,
alcohol, gliadin, chronic systemic
inflammation, Western diet (high fat,
low fiber), stress, enteric pathogens

Disruption of epithelial
barrier, increased permeability

Gut lumen

Intestinal
epithelium

Capillary

Tight junctions

Microbes/
pathogens

Bacterial
endotoxins
(LPS)

Antigenic
proteins &
peptides

Other antigens:
polysaccharides,
lipids, nucleic
acids

Paracellular
transport

Transcellular
transport

Some ions, small
compounds, nutrients,
microbes, & other
macromolecules enter via
transcellular mechanisms

Dysregulatory Immune Response Resulting in Systemic
Inflammation in Disease

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Functional Medicine

Long Description: Dysregulatory Immune Response

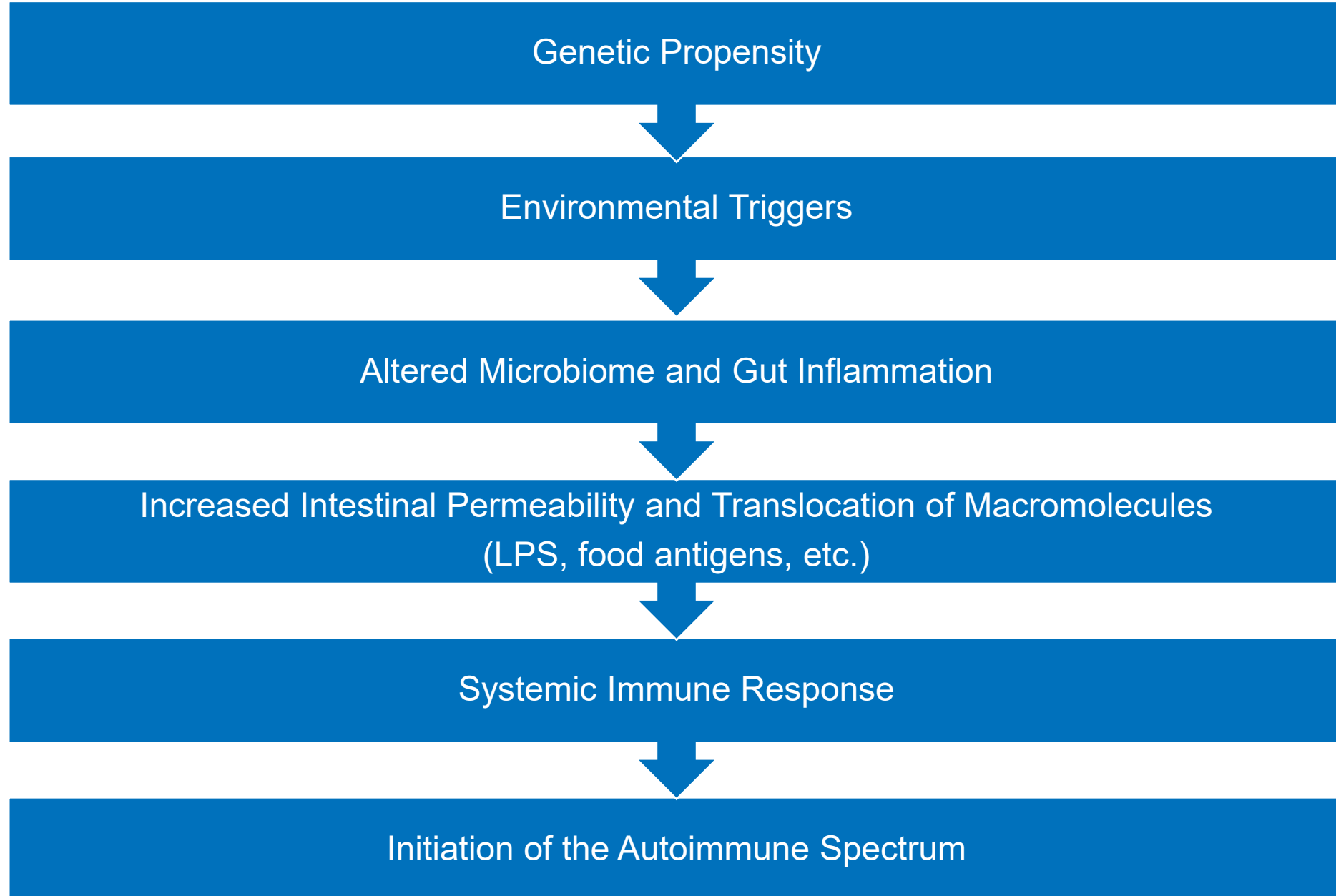
- N S A I Ds, dysbiosis, enterotoxins, alcohol, gliadin, chronic systemic inflammation, stress, enteric pathogens, and a Western diet high in fat and low in fiber can disrupt the epithelial barrier, increasing intestinal permeability. This allows pathogens, antigenic proteins and peptides, L P S, and other antigens including polysaccharides, lipids, and nucleic acids to cross through paracellular transport. Meanwhile, ions, small compounds, nutrients, microbes, and other macromolecules may enter through transcellular pathways. The resulting barrier breach triggers a dysregulated immune response, contributing to systemic inflammation and disease.

References

COMPONENTS OF INTESTINAL PERMEABILITY & MEDIATORS OF INCREASED IP

- De Munck TJI, Xu P, Verwijs HJA, et al. Intestinal permeability in human nonalcoholic fatty liver disease: a systematic review and meta-analysis. *Liver Int.* 2020;40(12):2906-2916. doi:10.1111/liv.14696
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Increased Intestinal Permeability and Autoimmunity



Clinical Focus

- Clinical findings of increased intestinal permeability.



Intestinal Permeability:

Representative Patient Clinical Findings

Food Sensitivities

Headaches

Brain Fog

Constipation

Fatigue

Diarrhea

Gas & Bloating

Joint pain



Intestinal Permeability

Chronic Conditions Associated With Increased Intestinal Permeability

Gastrointestinal	Systemic
Irritable bowel syndrome	Allergies and asthma
Inflammatory bowel disease	Infections
Celiac disease	Autoimmune conditions
Some cancers	Liver diseases
	Diabetes/metabolic conditions
	Mood disorders
	Neurological conditions
	Chronic pain
	Skin conditions
	Chronic fatigue syndrome

References

CHRONIC CONDITIONS ASSOCIATED WITH INCREASED INTESTINAL PERMEABILITY

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Most Common Clinical Findings of Increased Intestinal Permeability

Signs & Symptoms	
Gastrointestinal	• Often asymptomatic • Variable if present • Bloating • Gas • Abdominal pain • Diarrhea
Extraintestinal	• Fatigue • Headaches • Food sensitivities • Skin manifestations • Chronic pain • Joint pain • Mood changes

References

MOST COMMON CLINICAL INDICATORS OF INTESTINAL PERMEABILITY

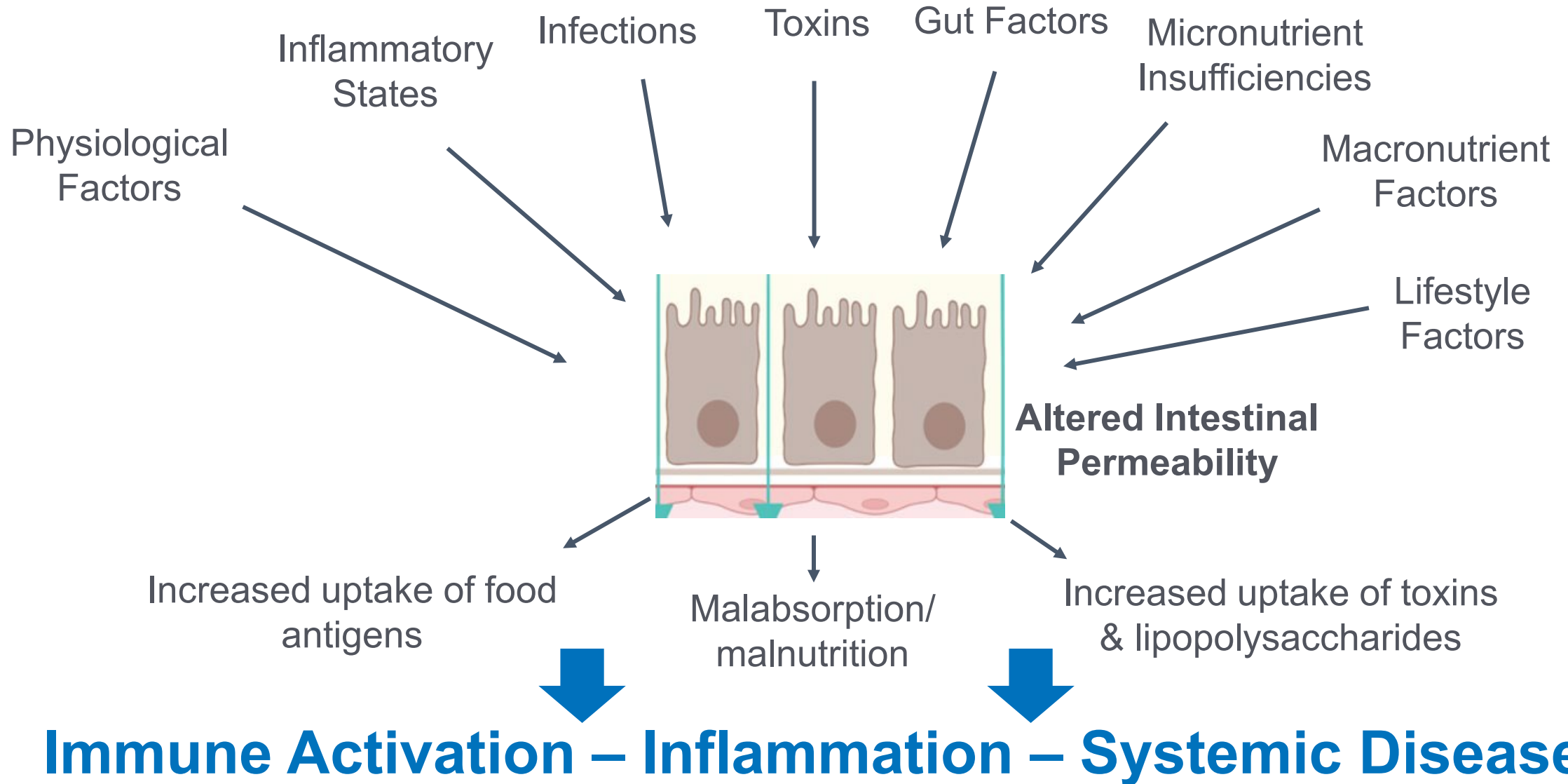
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Clinical Focus

- Antecedents, triggers, and mediators of increased intestinal permeability.



ATMs of Increased Intestinal Permeability



Long Description: A T Ms of Increased Intestinal Permeability

- Antecedents, triggers, and mediators of increased intestinal permeability include physiological factors, inflammatory states, dysbiosis, infections, toxins, gut factors, nutritional insufficiencies, and lifestyle factors. The resulting altered intestinal permeability leads to increased uptake of food antigens, malabsorption, malnutrition, and increased uptake of toxins and L P S, ultimately resulting in immune activation, inflammation, and systemic disease.

ATMs of Increased Intestinal Permeability

Physiological Factors

- Hypochlorhydria
- Constipation
- Diarrhea
- Intestinal hypoperfusion

Inflammatory States

- Autoimmune conditions
- Obstructive sleep apnea
- Food allergies
- Reaction to gliadin protein
- Concussion (TBI)

Infections

- Gastroenteritis

Toxins

- Alcohol
- Additives
- Cadmium
- Medications

ATMs of Increased Intestinal Permeability

Gut Factors

- Maldigestion
- Malabsorption
- Gut inflammation
- ENS dysfunction
- Dysbiosis

Micronutrient Insufficiencies

- Zinc
- Vitamin A
- Vitamin D
- Glutamine
- Arginine

Macronutrient Factors

- High-fat diet
- Simple sugars
- Refined carbs

Lifestyle Factors

- Overexercise
- HIIT
- Night-shift work
- Stress/elevated cortisol

References

ATMS OF INCREASED INTESTINAL PERMEABILITY

Nutrient Insufficiencies (Macronutrient & Micronutrient):

- Farré R, Fiorani M, Abdu Rahiman S, Matteoli G. Intestinal permeability, inflammation and the role of nutrients. *Nutrients*. 2020;12(4):1185. doi:10.3390/nu12041185
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Dysbiosis:

- Brown K, DeCoffe D, Molcan E, Gibson DL. Diet induced dysbiosis of the intestinal microbiota and the effects on immunity and disease. *Nutrients*. 2012;4(8):1095-1119. doi:10.3390/nu4081095
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Lifestyle Factors:

- Kelly JR, Kennedy PJ, Cryan JF, Dinan TG, Clarke G, Hyland NP. Breaking down the barriers: the gut microbiome, intestinal permeability and stress related psychiatric disorders. *Front Cell Neurosci*. 2015;9:392. doi:10.3389/fncel.2015.00392
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ATMS OF INCREASED INTESTINAL PERMEABILITY

Infections:

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Gut Factors:

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Toxins:

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ATMS OF INCREASED INTESTINAL PERMEABILITY

Inflammatory States:

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- Barceló A, Esquinas C, Robles J, et al. Gut epithelial barrier markers in patients with obstructive sleep apnea. *Sleep Med*. 2016;26:12-15. doi:10.1016/j.sleep.2016.01.019
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References

ATMS OF INCREASED INTESTINAL PERMEABILITY

Physiological Factors:

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Common Medications

Representative Example of Mediators of Increased IP

- NSAIDs
- PPIs
- Oral contraceptives
- Antibiotics
- Chemotherapies



References

MEDICATIONS AS MEDIATORS OF INTESTINAL PERMEABILITY

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Healing Leaky Gut – Autoimmune Connection



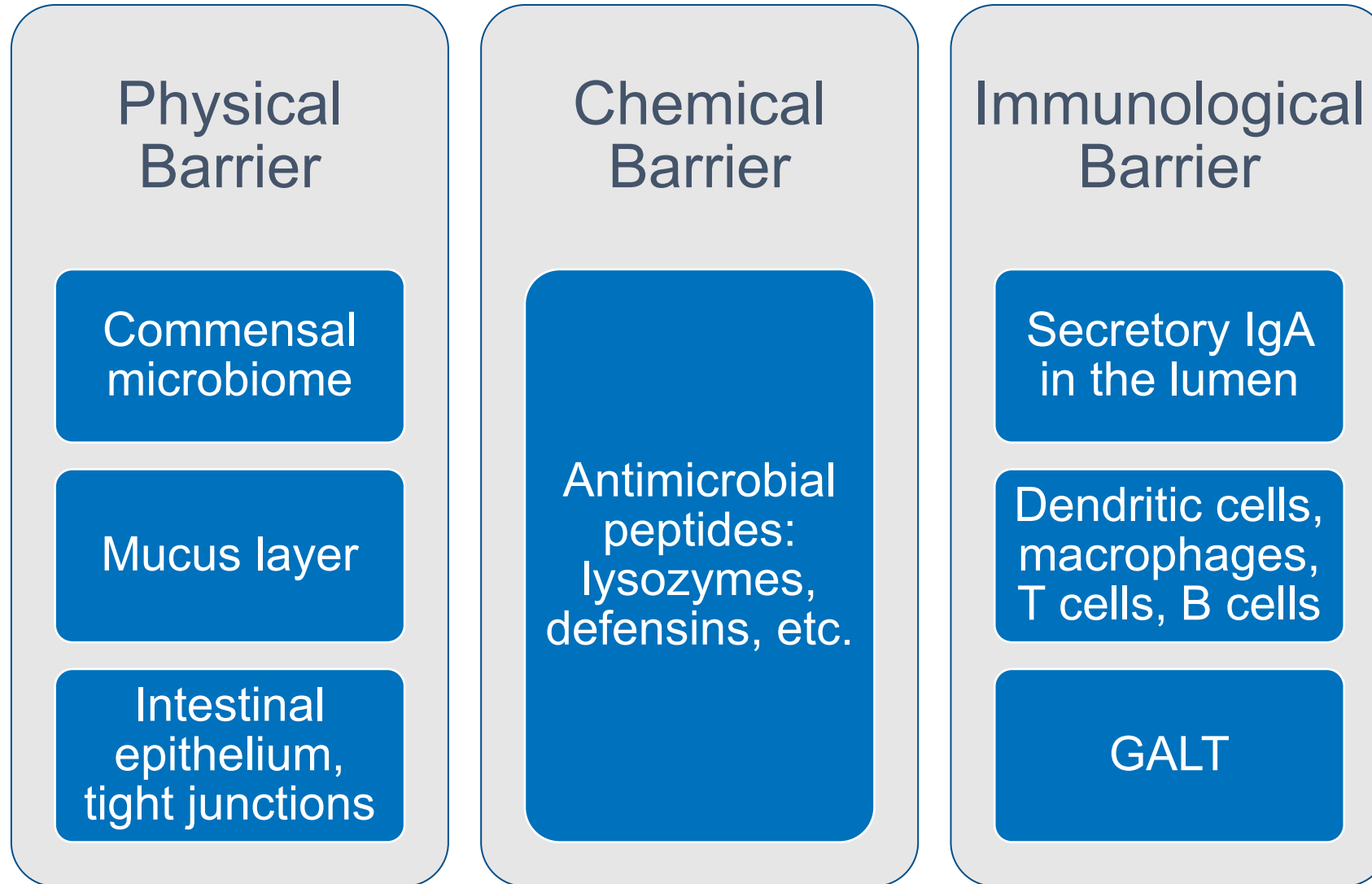
Image Courtesy of Swathi Rao, PA-C

Clinical Focus

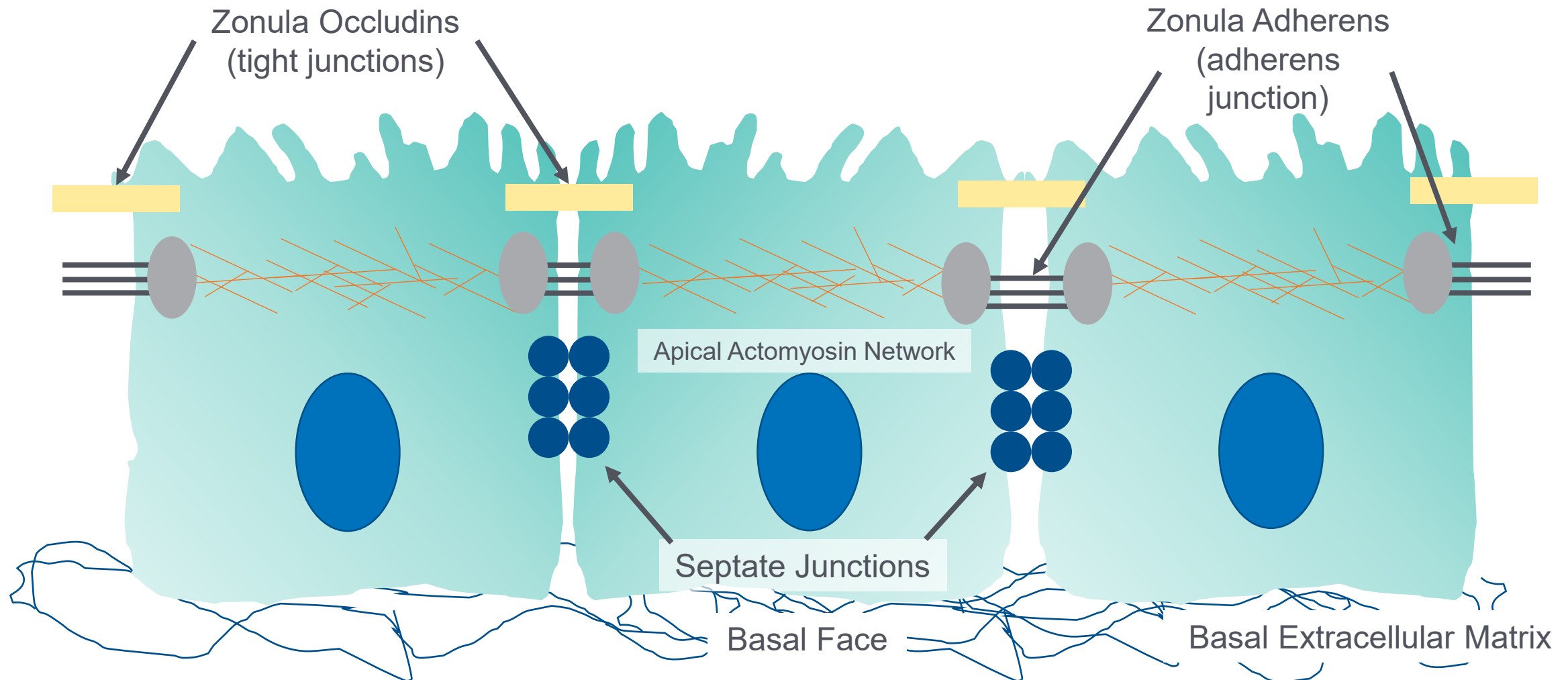
- Summary, conclusion, and key clinical takeaways.



Components of the Intestinal Barrier



Components of Normal Intestinal Permeability



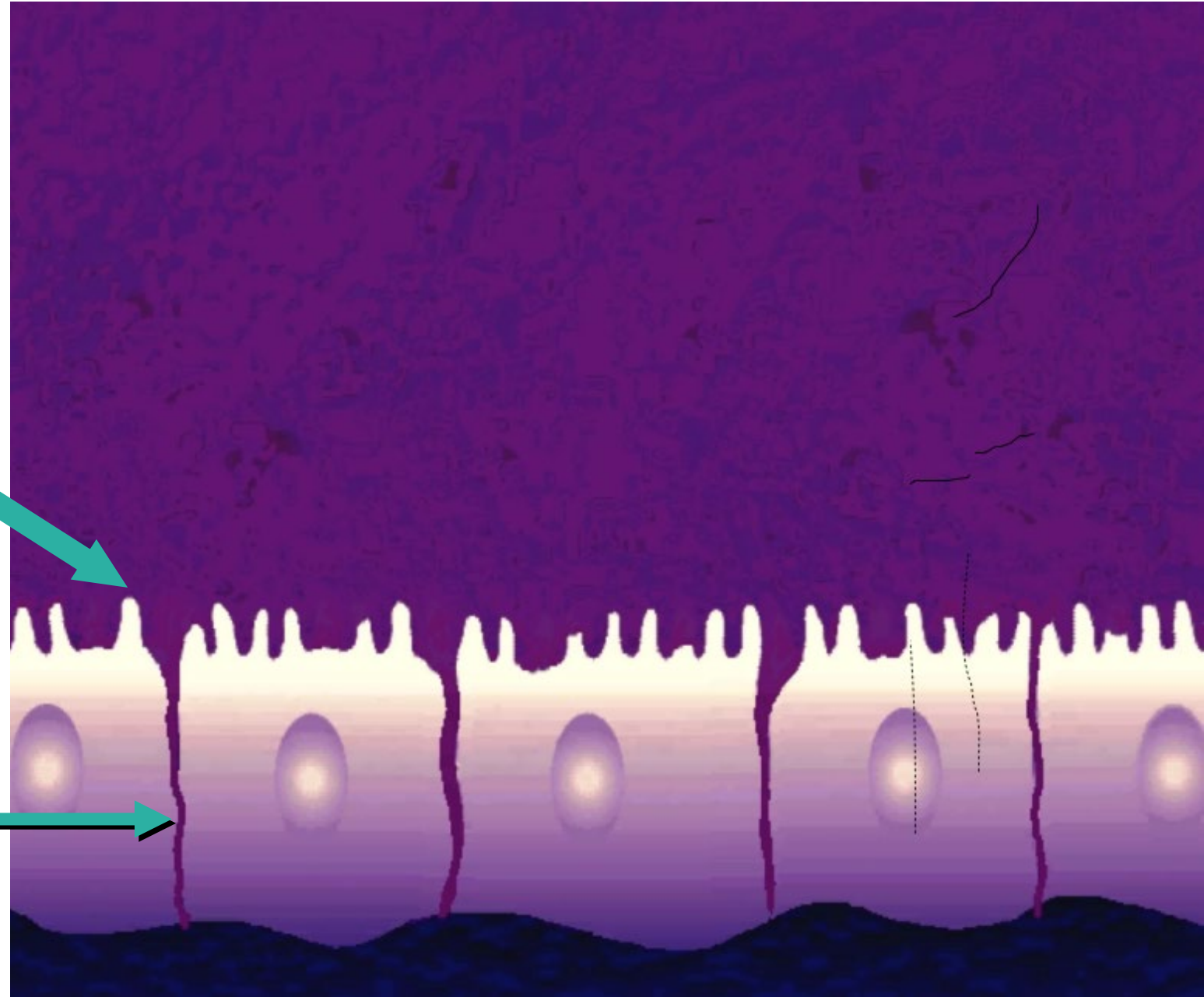
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- Epithelial cells are connected by zonula occludens, or tight junctions, at the apical surface. Beneath them are the zonula adherens, or the adherens junctions, connecting to the apical actomyosin network.

Leaky Gut & Malabsorption

**Damaged
Villi/Poor
Absorption**

**Damaged
Cell
Junctions**



Long Description

- Damaged villi and damaged cell junctions permit intestinal permeability and malabsorption.

NSAIDs, dysbiosis, enterotoxins, alcohol, gliadin, chronic systemic inflammation, Western diet (high fat, low fiber), stress, enteric pathogens

Disruption of epithelial barrier, increased permeability

Gut lumen

Intestinal epithelium

Capillary

Tight junctions

Microbes/
pathogens

Bacterial
endotoxins
(LPS)

Antigenic
proteins &
peptides

Other antigens:
polysaccharides,
lipids, nucleic
acids

Paracellular
transport

Transcellular
transport

Some ions, small
compounds, nutrients,
microbes, & other
macromolecules enter via
transcellular mechanisms

Dysregulatory Immune Response Resulting in Systemic
Inflammation in Disease

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Functional Medicine

Long Description: Dysregulatory Immune Response

- NSAIDs, dysbiosis, enterotoxins, alcohol, gliadin, chronic systemic inflammation, stress, enteric pathogens, and a Western diet high in fat and low in fiber can disrupt the epithelial barrier, increasing intestinal permeability. This allows pathogens, antigenic proteins and peptides, LPS, and other antigens including polysaccharides, lipids, and nucleic acids to cross through paracellular transport. Meanwhile, ions, small compounds, nutrients, microbes, and other macromolecules may enter through transcellular pathways. The resulting barrier breach triggers a dysregulated immune response, contributing to systemic inflammation and disease.

Most Common Clinical Findings of Increased Intestinal Permeability

Signs & Symptoms	
Gastrointestinal	• Often asymptomatic • Variable if present • Bloating • Gas • Abdominal pain • Diarrhea
Extraintestinal	• Fatigue • Headaches • Food sensitivities • Skin manifestations • Chronic pain • Joint pain • Mood changes

ATMs of Increased Intestinal Permeability

Physiological Factors

- Hypochlorhydria
- Constipation
- Diarrhea
- Intestinal hypoperfusion

Inflammatory States

- Autoimmune conditions
- Obstructive sleep apnea
- Food allergies
- Reaction to gliadin protein
- Concussion (TBI)

Infections

- Gastroenteritis

Toxins

- Alcohol
- Additives
- Cadmium
- Medications

ATMs of Increased Intestinal Permeability

Gut Factors

- Maldigestion
- Malabsorption
- Gut inflammation
- ENS dysfunction
- Dysbiosis

Micronutrient Insufficiencies

- Zinc
- Vitamin A
- Vitamin D
- Glutamine
- Arginine

Macronutrient Factors

- High-fat diet
- Simple sugars
- Refined carbs

Lifestyle Factors

- Overexercise
- HIIT
- Night-shift work
- Stress/elevated cortisol

Intestinal Permeability:

Manifestations – Mediators – Mitigators

APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE